Sounding Like Which AI? Machine-Style Overlap of arXiv Abstracts with Language Models of Different Vintages

Natik Chhabra
Independent researcher, Bilaspur, India
ORCID 0009-0004-8786-7703
https://github.com/NatikChhabra/Sounding-Like-Which-AI-

Abstract

Studies of LLM-assisted writing estimate how much text was produced with LLM help. We ask how much writing, by any author, falls on the LLM side of a stylistic boundary, and which LLMs define that side. We define **Machine-Style Overlap** (MSO): the share of documents that a calibrated classifier on content-word-free style features judges at least as likely to be LLM output as pre-ChatGPT human writing. In 131,448 arXiv abstracts (2015–2026), MSO was 9.2% before ChatGPT and by June 2025 stood 25.4 percentage points above its pre-ChatGPT trend, well above any placebo or instrument-drift gap. Recent LLMs learn from recent writing, so overlap with them is ambiguous; we compare reference generators by vintage. Against generators whose pretraining data predate ChatGPT, the gap is smaller but clear (7.5 points; 11.9 measured nonparametrically). With one generator at a time as reference, overlap with all but the newest rises and then falls, peaking later for later generators (exploratory rank correlation 0.79, $n = 18$). If this pattern persists, a fixed reference set, including a list of marker words such as *delve*, will eventually show the shift receding. In student essays, English learners' writing is less machine-like than their peers'. A GPT-2 perplexity instrument agrees, although it reproduces a published bias against non-native TOEFL essays.

1 Introduction

Corpus studies find traces of LLM-assisted writing in abstracts, peer reviews, press releases and encyclopedias (Kobak et al., 2025; Liang et al., 2024a,b, 2025; Brooks et al., 2024), and LLM-associated words have become more frequent in unscripted speech (Yakura et al., 2026; Anderson et al., 2025). Judging individual documents is harder: readers are poor at telling human from machine text (Clark et al., 2021; Jakesch et al., 2023; Casal and Kessler, 2023), detectors can be evaded (Sadasivan et al., 2025; Krishna et al., 2023; Dugan et al., 2024), and some detectors have flagged non-native writers' human text as AI-generated at high rates (Liang et al., 2023).

Existing measurements answer a *prevalence* question: what fraction of a corpus was produced or modified with an LLM? We ask an *overlap* question: what fraction of a genre's writing, by any author, is stylistically closer to LLM output than to pre-ChatGPT human writing? The two differ in how they handle text written before ChatGPT. Prevalence estimators report a small "LLM share" for it, which studies read as consistent with false positives (Liang et al., 2024b), subtract as baseline misclassification (Lin and Zhu, 2025), or use to calibrate thresholds (Brooks et al., 2024). For a fixed feature map and reference set, we regard that share as a property of the two text distributions and study how it changes over time.

Both questions share a complication that prevalence studies seldom discuss: "LLM style" is defined by whichever models serve as the reference. Recent models are trained on recent text, so a rising overlap with them may mean that writing moved towards the models or that the models learned from the writing. We therefore compare reference generators by *vintage* on a fixed task: models whose pretraining data predate ChatGPT, models released in 2023–2025, and models released in 2026.

Contributions. (1) We define Machine-Style Overlap (MSO), a classify-and-count quantity (Forman, 2008) that, for an optimal classifier, equals the Bayes classifier's false-positive rate on pre-ChatGPT writing; we estimate it with calibrated classifiers and report how the estimate varies across them (§3). (2) We measure it in arXiv abstracts against 19 reference generators released from 2023 to 2026, including four whose pretraining data predate ChatGPT and four released in

2026 (§§4–5). (3) Abstracts moved far into the LLM region after 2022, more in computer science than in mathematics. Placebo dates and instruments trained on earlier years give gaps at most about a quarter as large (§6.3). (4) The shift is present, though smaller, against generators that are unlikely to have learned it from post-ChatGPT writing; overlap with a single generator tends to peak later for more recent generators, so the trajectory depends on the reference's vintage (§6.4). (5) Secondary results (§§6.5–6.6): words mined from each generator rose until 2025 and mostly fell in 2026; stylistic dispersion did not change beyond its trend once length is fixed; parliamentary speech changed little; and, contrary to our plan, English learners' essays are *less* machine-like than their peers'. Our analysis plan was committed to version control before any result was computed (Appendix A). It stated five hypotheses: pre-ChatGPT MSO is non-zero and differs by field (H1); arXiv MSO rises above its pre-ChatGPT trend (H2), more than in a less-exposed genre (H3) and more in computer science than in mathematics (H4); and learner essays have higher MSO (H5). The vintage analysis was added after simulated review.

2 Related Work

Prevalence of LLM-assisted writing. Excess-vocabulary methods compare word frequencies with a counterfactual extrapolated from pre-LLM years (Kobak et al., 2025; Holzwarth et al., 2026; Lee, 2026); distributional methods fit a mixture of human and LLM word distributions (Liang et al., 2024a,b, 2025; Geng and Trotta, 2024); classifier- and detector-based studies label documents (Lin and Zhu, 2025; Brooks et al., 2024; Sun et al., 2025b; Khan et al., 2026). All target the share of text produced with LLM help; LLM adoption also changes how many papers scientists produce (Kusumegi et al., 2025). Closest to ours, Bentley et al. (2026) track a composite LLM-associated style index alongside vocabulary-complexity measures in arXiv abstracts, and Askitas (2025) finds an abrupt stylometric shift in economics working-paper abstracts in 2023 with an event-study design and a classifier. Chang (2026) shows that PubMed, Wikipedia and Stack Exchange text became more predictable to small language models over 2014–2024, a trend that began before generative AI; our pre-ChatGPT slope points the same way. Our estimand differs: a thresholded overlap (the flag rate of a calibrated classifier) whose pre-ChatGPT level is itself of interest. We also exclude content words from the features, since authors react to publicised markers (Geng and Trotta, 2025), and compare reference generators by training-data vintage.

Homogenisation and human adoption. Writing or ideating with instruction-tuned models reduces content diversity across users (Padmakumar and He, 2024; Doshi and Hauser, 2024; Anderson et al., 2024), and LLM rewriting compresses stylistic variance (Sourati et al., 2026). Preference tuning reduces output diversity (Kirk et al., 2024; Wu et al., 2025) and plausibly produces LLM-overused words (Juzek and Ward, 2025). LLM-associated vocabulary has grown in human speech (Yakura et al., 2026; Anderson et al., 2025), academic authors adapt their LLM use once overused words are publicised (Geng and Trotta, 2025), and adoption of LLM revision varies across disciplines and author groups (Lin and Zhu, 2025). Models trained on their own outputs degrade when synthetic data replace real data (Shumailov et al., 2024) but not when the two accumulate (Gerstgrasser et al., 2024). Our vintage comparison asks whether human text still provides a reference independent of LLM style.

Style of human and LLM text. LLM text differs systematically from human text in lexico-grammatical features (Reinhart et al., 2025; Muñoz-Ortiz et al., 2024; Berber Sardinha, 2024; Herbold et al., 2023) and in recurring part-of-speech templates (Shaib et al., 2024), but texts cluster by genre more than by source (Rallapalli et al., 2026), so any overlap measure must be genre-specific. Distribution-level comparisons (Pillutla et al., 2021; Yang et al., 2026) evaluate models against human text; we track where human documents fall relative to an LLM region.

Detection and perception. Readers misjudge machine text (Clark et al., 2021; Jakesch et al., 2023; Casal and Kessler, 2023; Gao et al., 2023); we collect no human judgements. Zero-shot detectors use token-likelihood statistics (Hans et al., 2024; Bao et al., 2024), and detection needs more text as machine and human distributions converge (Chakraborty et al., 2024); off-the-shelf detectors have flagged non-native writing (Liang et al., 2023), although detectors trained on a large writing assessment showed no such bias (Jiang

et al., 2024) and false-positive rates on non-native manuscripts vary widely (Park et al., 2026). Detectors transfer unevenly to unseen generators (Pu et al., 2023; Li et al., 2024), and individual models have distinctive, identifiable output patterns (Sun et al., 2025a), so a style region depends on the models that define it.

3 Machine-Style Overlap

Fix a genre and a feature map ϕ from documents to style vectors. Let P_{pre} be the distribution of $\phi(d)$ for human documents written before ChatGPT's release (30 November 2022; "pre-ChatGPT" throughout), and P_{LLM} the distribution for LLM output in the same genre, an equal-weight mixture over a reference set of generators and prompting styles chosen by us. For any population S of documents,

$$\text{MSO}(S) = \Pr_{d \sim S}\left[\, p_{\text{LLM}}(\phi(d)) \geq p_{\text{pre}}(\phi(d)) \,\right]. \tag{1}$$

With equal priors the Bayes posterior is $\eta(x) = p_{\text{LLM}}(x)/(p_{\text{LLM}}(x) + p_{\text{pre}}(x))$, so $\text{MSO}(S) = \Pr_S[\eta(\phi(d)) \geq \frac{1}{2}]$; we call $\{x : \eta(x) \geq \frac{1}{2}\}$ the *LLM region*. MSO is the flag rate of the Bayes-optimal classifier, that is, classify-and-count (Forman, 2008) with a calibrated threshold.

Pre-ChatGPT overlap. For $S = P_{\text{pre}}$, MSO is the Bayes classifier's false-positive rate, and the Bayes error is $\frac{1}{2}\big(\text{MSO}(P_{\text{pre}}) + \Pr_{\text{LLM}}[\eta < \frac{1}{2}]\big)$, which is zero only when the two styles are separable and reaches $\frac{1}{2}$ when they coincide. Prevalence studies that subtract a pre-period share as misclassification remove the analogue of this quantity under their own feature map.

Overlap after ChatGPT. For later populations S_t, $\text{MSO}(S_t)$ measures how much writing at time t falls in the region. It says nothing about authorship: a rise is consistent with undisclosed LLM use, with humans adopting LLM style, or both. If a fraction α of documents were LLM output and the rest unchanged human writing, $\text{MSO}(S_t) = \text{FPR} + \alpha(\text{TPR} - \text{FPR})$; we use this identity only to translate MSO into an implied α for comparison with prevalence studies.

Estimation. We call a trained, calibrated classifier with its 0.5 threshold an *instrument*. A finite instrument only approximates η, and its flag rate is neither an upper nor a lower bound on the Bayes quantity. We therefore report how MSO varies across classifier families, held-out generators, generator sets and thresholds; confidence intervals in the text are document-level bootstrap or regression intervals and exclude this instrument variation.

4 Data

arXiv abstracts. We sampled eight arXiv archives by primary category (cs, math, stat, eess, q-bio, econ, astro-ph, cond-mat; "fields" below). For each field and each month from January 2015 to September 2026, we fetched through the arXiv API all papers submitted in one randomly placed window of consecutive days and drew up to 200 at random. Field-months with at least 200 papers therefore carry equal weight. Smaller fields contribute fewer documents, especially in early years, so the field mix changes (eess made up 1.5% of scored abstracts in 2015 and 16.2% in 2022), and the overall series averages a stratified sample and does not represent the arXiv population. The API returns the latest version of each abstract, so papers first submitted before ChatGPT but revised after it are excluded (0.5% of sampled 2016 papers, 7.8% of 2021 and 24.2% of 2022 papers). Recent papers have had less time to be revised, but scoring only never-revised papers barely changes yearly MSO (2026: 44.7% vs 44.7%). After removing abstracts under 50 words, 131,448 abstracts remain.

Other human text. *Parliamentary speech:* Australian House of Representatives Hansard (Katz and Alexander, 2023, 2025), up to 150 spoken speeches per month from 2015 to July 2025 (13,725 speeches). *Student essays:* PERSUADE 2.0 (Crossley et al., 2024), 14,797 pre-ChatGPT essays by US students in grades 6–12 on 14 prompts, with English-language-learner (ELL) status and a holistic score. In every genre, half of the pre-ChatGPT documents (the *train pool*) train the instrument and seed LLM prompts; MSO is measured on the rest.

Reference generators. Table 1 lists them. Local models answered three prompt styles, 150 prompts each: *P1* write the text from a title or topic; *P2* improve the wording of a human draft (LLM-assisted revision); *P3* write the text and "avoid typical AI phrasing". Prompts were seeded only from train-pool documents; local models used temperature 0.8 and fixed seeds. The 2026 model (Claude Opus 5.5) answered the same

Generator	Released	Cutoff
RAID, 2023 chat models (Dugan et al., 2024)		
GPT-4 (gpt-4-0613)	2023	Sep 2021[a]
Llama-2-Chat 70B	2023	Sep 2022[b]
ChatGPT (3.5-turbo-0613)	2023	(2021)[c]
Mistral-7B-Instr. v0.1; MPT-30B-chat; Cohere command	2023	
Local; main instrument, all genres		
Mistral 7B Instruct v0.3	2024-05	
Llama 3.1 8B	2024-07	Dec 2023
Phi-4 14B	2024-12	Jun 2024
Gemma 3 12B	2025-03	Aug 2024
Granite 3.3 8B; Qwen3 8B	2025-04	
gpt-oss 20B	2025-08	Jun 2024
Sub-agent sessions; main instrument, all genres		
Claude Opus 5.5 (2026 model)	2026	
Local; vintage analysis only (abstracts, P1–P3)		
Llama-2-Chat 7B	2023-07	Sep 2022[b]
Vicuna 7B v1.5[d]	2023-08	Sep 2022[b]
Qwen3.5 9B	2026-Q1	
Gemma 4 12B	2026-06	Jan 2025
Granite 4.2 8B	2026-08	

Table 1: Reference generators, with release month and pretraining-data cutoff where the developer states one. [a]OpenAI (2023); [b]Touvron et al. (2023); [c]reported only in secondary sources, so ChatGPT enters the pre-ChatGPT-data region only as a sensitivity check; [d]fine-tuned from Llama 2 (LMSYS, 2023) with the Vicuna recipe of training on user-shared ChatGPT conversations (Zheng et al., 2023). Other dates: Jiang et al. (2023); Mistral AI (2024); Meta (2024); Microsoft (2024); Google DeepMind (2025); Qwen Team (2025); IBM Granite Team (2026); OpenAI (2025); Qwen Team (2026); Google DeepMind (2026b,a).

prompt instances through sub-agent sessions of 50 prompts each, without control over decoding; it is also the system that drafted this paper (see Acknowledgements). For abstracts we add the RAID generations without adversarial attack (Dugan et al., 2024), written from titles of arXiv papers by six 2023 chat models. The main instrument uses 14 generators, each capped at 450 training texts. Five further open models were added after review, for the vintage analysis only: two whose pretraining data predate ChatGPT and three released in 2026. The latter give the 2026 comparison a second source with logged decoding, and each pairs with a 2025 model of the same family. Hansard and essays use the seven local models and the 2026 model.

5 Method

Cleaning. All texts are normalised alike. We remove chat wrappers ("Here's the abstract:"), lines echoing the prompt's title, markdown and trailing notes, and map typographic variants and misdecoded characters to plain equivalents. LaTeX math becomes a placeholder and LaTeX dashes (`--`, `---`) become Unicode dashes: human arXiv abstracts write dashes in LaTeX and chat models in Unicode, a difference that would otherwise separate the sources by encoding alone. Texts are truncated to 250 words.

Features and instruments. Features exclude content words to limit topic sensitivity: per-token rates of closed-class words, coarse part-of-speech trigrams, dependency labels and punctuation marks, plus mean and standard deviation of sentence length, mean word length and MTLD lexical diversity (McCarthy and Jarvis, 2010), computed with spaCy 3.8 (`en_core_web_sm`). Features present in fewer than 1% of training documents are dropped (top features in Appendix B). The primary instrument is L2-regularised logistic regression (C=0.1) on standardised features with isotonic calibration; the second uses gradient-boosted trees, calibrated the same way. Classes are balanced, so a calibrated probability of 0.5 is a likelihood ratio of one. Cross-validation and calibration folds are grouped so that a revised text and its human source, and all speeches by one parliamentarian, never fall in different folds. For essays we add two instruments: logistic regression on sentence embeddings (all-MiniLM-L6-v2), and a one-feature isotonic classifier on mean token negative log-likelihood, the statistic behind perplexity-based detectors, under a small causal language model (Qwen2.5-0.5B, or GPT-2 small, whose training data predate the TOEFL and PERSUADE releases). Both are trained on the same human and LLM essays as the primary instrument.

Time series. We compute monthly MSO and fit an interrupted time-series model: weighted least squares with a level shift and a slope change at December 2022 and Newey–West (6 lags) standard errors. The *trend gap* is the fitted post-ChatGPT value minus the extrapolated pre-ChatGPT trend in June 2025, a month fixed in code before any time-series result. As robustness checks we vary the pre-trend (quadratic; start in 2017 or 2019; end in 2020), fit the post-ChatGPT line only up to June 2025, hold the field mix fixed, compute a nonparametric gap (mean observed January–June 2025 value minus the extrapolated trend), and bootstrap the gap with 12-month residual blocks.

Placebo breaks are placed every second month from June 2016 to December 2021, each evaluated 12 months later, and at two fixed dates (November 2019 and 2020) evaluated in June 2022. Drift placebos train the instrument only on human abstracts up to 2016, 2017 or 2018, place a break in the following January and test for a gap in June 2022, 41 to 65 months later and before any LLM writing assistance, so any gap they show reflects instrument ageing.

Vintage. To compare vintages on a fixed task, we train one instrument per generator on its title-only (P1) outputs (RAID outputs are title-based). A further instrument pools the generators whose pretraining data predate ChatGPT (the *pre-ChatGPT-data region*): GPT-4 and Llama-2-Chat 70B from RAID, plus Llama-2-Chat 7B and Vicuna 7B run locally on P1. Many RAID outputs summarise the paper ("In the paper titled…"); we drop those matching a summary pattern from this region and from each RAID generator's own region (5.4% of the RAID part of this set; up to 32.9% for Cohere's model).

Marker words. We track two lexical series: the share of abstracts containing a published marker word (*delve*, *underscore*, *showcase*, *meticulous*, *intricate*, *intricacies*, *commendable*, *boast*; Kobak et al., 2025; Gray, 2024; Yakura et al., 2026), and, for each reference set, the share containing one of 20 words mined from that set's own outputs: the words with the highest smoothed log ratio of document frequency between the outputs and the human abstracts their prompts were built from, so that most topic words of the prompt set cancel.

6 Results

6.1 Separability of human and LLM text

The instruments separate pre-ChatGPT human text from LLM text well (cross-validated AUC 0.939 for abstracts, 0.968 for speech, 0.988 for essays). Generators held out of training are mostly recognised (arXiv recall from 59.6% for Cohere to 95.6% for GPT-4; Table 4 in Appendix C), but the 2026 model is not: only 20.7% of its abstracts and 36.4% of its essays fall in a region built without it. Its revisions of human drafts are also hard to separate from human text (AUC 0.675). Lexical diversity is the top LLM-typical feature; human abstracts use more coordination, adjectival modifiers and longer sentences (Appendix B).

6.2 Pre-ChatGPT overlap varies by field (H1)

Before ChatGPT, 9.2% (95% CI 8.9–9.5) of arXiv abstracts fell in the LLM region, as did 5.4% (4.9–6.1) of speeches and 4.0% (3.6–4.5) of student essays. The arXiv baseline varies by field (Figure 3 in Appendix E; $\chi^2(7) = 354.55$, $p<0.001$), from 4.4% in astrophysics to 15.3% in economics. Fields also differ in abstract length; reweighting each field to the overall length distribution changes the values to 6.1% and 14.0%, and abstracts without math give a similar ordering. Overall MSO was also rising slowly before ChatGPT (0.5 pp per year).

6.3 Abstracts moved into the region (H2, H4)

Yearly arXiv MSO ranged from 7.3% to 10.2% in 2015–2021, was 11.1% in 2022, then 17.7%, 27.8% and 40.6% in 2023–2025 and 44.7% in January–September 2026. The monthly series has been roughly flat since late 2025 (Figure 1a). The trend gap in June 2025 is 25.4 pp (95% CI 23.2–27.7; block bootstrap 23.3–27.3), about 2.1 times the extrapolated trend value. Across the variants in Table 2 that change the threshold, the sample or the trend, the gap stays between 23.5 pp (threshold 0.6) and 29.1 pp (abstracts without math). It is smaller when the generators that write closest to human abstracts are left out of the region: without the 2026 model (22.2 pp) or without revisions of human drafts (17.9 pp). The gap does not rest on one feature family. Removing MTLD, the strongest feature, gives 25.4 pp, and removing all closed-class words, POS trigrams, dependency labels, punctuation or sentence statistics gives between 22.9 and 26.6 pp. The analytic Fast-DetectGPT criterion (Bao et al., 2024), computed with GPT-2 small, separates our human and LLM abstracts only moderately (AUC 0.737), and its flag rate on 600 abstracts per year does not rise (8.3% in 2015, 5.3% in 2026); the original method's larger scoring models may not miss the shift. Of the 34 placebo breaks, 8.8% give a gap with $p < 0.05$, close to the nominal 5%, and the largest gap is 3.2 pp. At the real gap's 30-month horizon, 8.3% give $p < 0.05$ and the largest is 5.8 pp. The three drift placebos give gaps of 6.5, -2.3 and -1.0 pp (Table 2). The positive one comes from the instrument trained on 2015–2016 only, extrapolated over 65 months from a two-year trend. Instrument ageing can thus produce gaps of a few points, up to about a quarter of

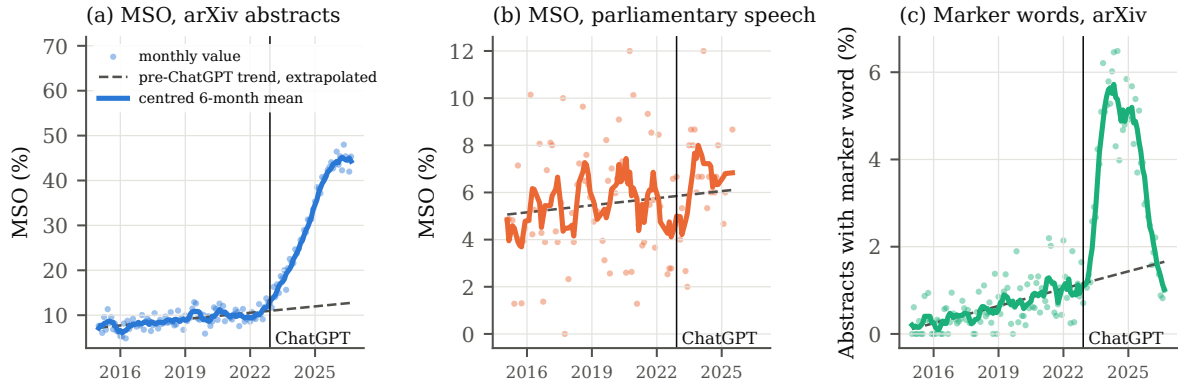


Figure 1: Monthly MSO for (a) arXiv abstracts and (b) Australian parliamentary speech (different y-scales), and (c) the share of arXiv abstracts containing a published marker word. Dashed: pre-ChatGPT linear trend, extrapolated. Series (a) rises well above trend, (b) shows a small gap and (c) peaks in 2024.

the observed gap. Yearly residuals from the pre-ChatGPT trend stay within ±1 pp through 2022 and reach 28.5 pp in 2025. *Fields (H4).* The gap is largest in computer science (38.1 pp) and quantitative biology (36.2) and smallest in mathematics (10.5) and astrophysics (13.7). Computer science exceeds mathematics by 27.6 pp (CI 22.2–32.9), and all eight field gaps remain significant after Holm correction (all adjusted $p<0.001$). If nothing else had changed, the mixture identity (§3) would imply that 42.1% of computer-science and 7.8% of mathematics abstracts were LLM output in 2024 (44.4% and 8.2% using held-out-generator recall). Liang et al. (2024b) estimate 17.5% and 4.9% of abstract sentences by February 2024, the same ordering at lower levels. Implied shares are larger if partly revised documents enter the region or authors adopt LLM style without an LLM, which the identity cannot tell apart; lower recall in mathematics (75.8% vs 85.8%) would raise its share.

Variant	Gap (pp)	95% CI or p
Primary (linear trend, ITS)	25.4	23.2–27.7
Threshold 0.4 / 0.6	27.2 / 23.5	
Abstracts without math	29.1	26.4–31.8
No LLM-topic abstracts	23.9	21.9–25.9
Length-reweighted	24.2	22.0–26.3
No dash features	26.1	23.9–28.3
Without the 2026 model	22.2	18.7–25.8
Without revision (P2) texts	17.9	15.7–20.2
Equal field weights	26.7	24.0–29.3
Pre-ChatGPT field weights	24.4	22.3–26.5
Post-trend fitted to June 2025	27.5	25.6–29.3
Nonparametric, linear trend	26.3	
quadratic trend	26.9	
trend from 2017	26.5	
trend from 2019	27.4	
trend to 2020	25.4	
Placebos (no LLM writing assistance; gap in June 2022)		
Break Nov 2019	−1.1	$p = 0.258$
Break Nov 2020	−0.8	$p = 0.335$
Drift, instrument to 2016	6.5	$p = 0.061$
Drift, instrument to 2017	−2.3	$p = 0.020$
Drift, instrument to 2018	−1.0	$p = 0.158$

Table 2: arXiv trend gap in June 2025 (percentage points above the extrapolated pre-ChatGPT trend) under robustness variants. Nonparametric rows: mean January–June 2025 value minus the trend.

6.4 Sounding like which AI?

Generators whose pretraining data predate ChatGPT. The pre-ChatGPT-data region (GPT-4, Llama-2-Chat 70B, Llama-2-Chat 7B and Vicuna 7B on the title-only task) reflects models unlikely to have learned their style from post-ChatGPT scientific writing. Their pretraining data predate ChatGPT, and their 2023 instruction tuning used annotator data or shared ChatGPT conversations and, as far as is documented, no recent abstracts. Abstracts still moved into it: MSO was 6.5% in 2022, 15.5% in 2024 and 17.9% in 2025, and the trend gap is 7.5 pp (CI 4.6–10.4), against 20.3 pp for a title-only region built from the 2024–2025 open models (Figure 2a). Its subsets point the same way but differ in size: the RAID models alone give 8.6 pp with summary-style outputs kept and 7.4 pp with ChatGPT added, and the two small local models, run on exactly our prompts, give 3.5 pp (CI 1.5–5.5). Three quarters of the region's training texts come from RAID, so its generators are not equally weighted and the pooled estimate leans on RAID's prompt. Part of the shift is therefore towards the style of

the pre-ChatGPT-data generators and is unlikely to be an artefact of newer models learning from newer abstracts, though the estimate from our own prompts is small. Drift placebos for this region (instruments trained on abstracts up to 2016, 2017 or 2018) give gaps of 0.5, −1.3 and −2.0 pp. In 2026 its MSO fell back to 11.9%. Because overlap with these generators peaked in 2025, a straight post-ChatGPT line fitted through 2026 understates their June 2025 gap. Measured nonparametrically, the gap is 11.9 pp against 22.1 pp for the 2024–2025 open models, and 6.6 pp for the two small local models.

One generator at a time. With the region defined by a single generator on the title-only task, MSO rose from 2022 to 2025 under every generator except Cohere's chat model. Quarter by quarter, overlap with each generator but the 2026 model rose and then turned down, and the turn tended to come later for later generators (Figure 2b). It peaked between 2024Q3 and 2025Q3 under the 2023 models other than Cohere's, in 2025Q2–Q3 under the 2024 models, in 2025Q4 under Gemma 3, Granite 3.3 and Qwen3, and in 2026Q1–Q2 under gpt-oss, Qwen3.5 and Granite 4.2. Under the 2026 model it was still rising in 2026Q3. The rank correlation between release date and peak quarter is 0.79 ($n = 18$, $p<0.001$; the 2026 model is excluded because its release month is not used). All generators score the same abstracts, so their errors are correlated; a document bootstrap within quarters gives 0.64–0.86, and without the 2023 models the correlation is 0.82 ($n = 10$). Individual peak quarters are less certain, typically by one or two quarters (Appendix D). The analysis is exploratory, chosen after a first test: the rank correlation of release date with the 2025-to-2026 change is 0.67 on annual means but 0.18 ($p = 0.464$) from 2025Q4. All three 2026 open models peaked before or in the quarter of their release (Gemma 4 12B, released in June 2026, in 2025Q4), so authors' use of them cannot explain the timing. On annual means, MSO fell from 2025 to 2026 under 15 of the 19 generators and rose under gpt-oss (29.4% to 38.5%), Qwen3.5, Granite 4.2 and the 2026 model (29.3% to 39.2%). Each change has a document-bootstrap interval that excludes zero and keeps its sign across five draws of the human training sample and ten refits on 80% of the generator's texts (Appendix D), except under Gemma 4 (−0.9 pp, CI −1.8 to 0.1). Measured from 2025Q4 instead, only gpt-oss, Granite 4.2, the 2026 model and, barely, Qwen3.5 (+0.4 pp) rose. In all three same-family pairs, the newer model's change was higher. From 2026Q2 to 2026Q3, overlap fell under all 18 generators but the 2026 model; for gpt-oss and Granite 4.2 that single quarter is the whole downturn. On the 9 generators with a stated pretraining cutoff, cutoff and release date order the peaks about equally well (rank correlations 0.59 and 0.62), so we cannot separate the recency of post-training from that of pretraining data. A region pooling the three 2026 open models, run with logged decoding, also rose (28.9% to 34.9% on annual means), so the late peak does not rest on the uncontrolled generator.

The aggregate 14-generator series (Figure 1a) still rose in 2026, carried by gpt-oss and the 2026 model, although 12 of its 14 single-generator regions fell. If overlap keeps turning down later for later generators, a fixed reference set will eventually show the shift receding, even if writing keeps moving. Of our comparisons, only the pre-ChatGPT-data region is free of the question of who learned from whom.

6.5 Marker words

The share of abstracts with a published marker word rose from 1.1% in 2022 to 5.3% in 2024 and fell to 1.5% in 2026 (Figure 1c); its June 2025 trend gap was 1.9 pp (CI 0.8–3.0). These words come from 2023–2024 models, so we also compare with words mined from the generators that define each style region. For every generator, the share of abstracts using one of the 20 words mined from its own outputs (§5) rose from 2022 to 2025 (for the pre-ChatGPT-data generators, from 17.4% to 34.9%), as did style overlap under every generator except Cohere's chat model. In 2026 the share fell for 17 of the 19 generators, including gpt-oss and Qwen3.5, under which style overlap rose. It kept rising only for Granite 4.2 (33.6% to 35.6%) and the 2026 model (49.4% to 57.4%), whose mined words are ordinary academic vocabulary. Under gpt-oss and Qwen3.5 the mined-marker share fell in 2026 while style overlap rose, so a falling marker share does not by itself show that writing stopped moving. Quarter by quarter, marker and style peaks follow no consistent order: the marker share peaked earlier than style overlap for 7 generators, later for 6 and in the same quarter for 5. At the document level the published markers and MSO pick out largely different abstracts (point-

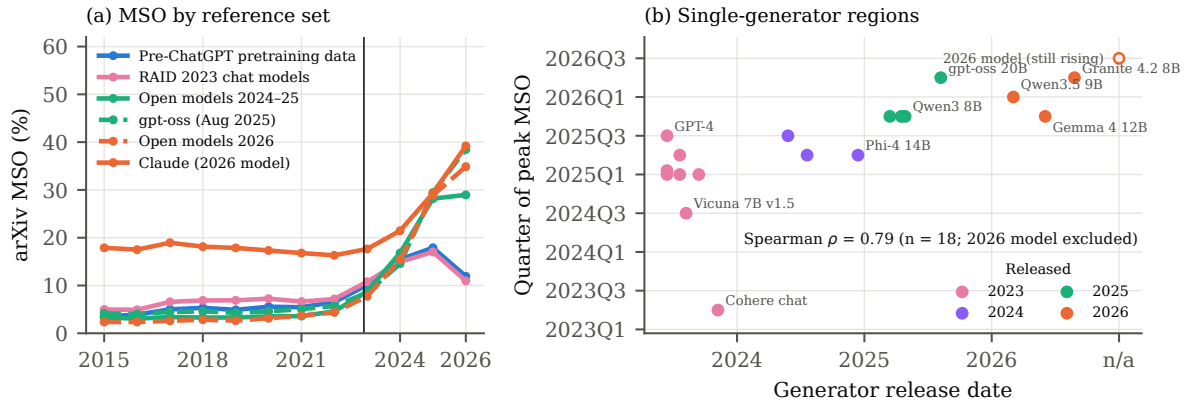


Figure 2: (a) Yearly arXiv MSO with the region defined by generators of different vintages (title-only prompts). (b) Quarter in which MSO peaked when the region is defined by one generator at a time, against that generator's release date (open circle: the 2026 model, whose release month is not used; still rising in the last quarter). Overlap with later generators tends to peak later.

biserial $r = 0.12$; 5.9% of in-region vs 1.4% of other abstracts contain one).

6.6 Secondary results

Parliamentary speech (H3). The Hansard trend gap is small (1.3 pp, CI −0.7 to 3.4), and the arXiv gap exceeds it by 30.7 pp (CI 28.3–33.2) with both regions built from the same generators. Hansard is a weak comparison: it fails its own placebo grid (23.5% of breaks with $p < 0.05$) and its gap depends on the pre-trend specification (Appendix F). *Dispersion.* With abstracts cut to 150 words, the stylistic spread of abstracts did not fall faster after ChatGPT than before (2025 gap 0.05, CI −0.36 to 0.51; Appendix G). *Learner essays (H5, not supported).* Our plan predicted higher MSO for English learners, but it is lower (2.3% vs 4.2%; odds ratio 0.55, 0.87 after adjusting for proficiency), and sentence-embedding and perplexity instruments agree in direction. A GPT-2 perplexity instrument flags 63.7% of TOEFL essays but 8.0% of US 8th-grade essays, reproducing the bias reported by Liang et al. (2023), yet it flags the PERSUADE learners less often than their peers, because their writing is less predictable (Appendix H).

7 Discussion

Whether abstracts were still moving towards LLM style in 2026 depends on the reference LLM. Overlap tended to peak later for more recent generators, and under three of the four 2026 generators it peaked in 2026 or was still rising. Words mined from most generators fell in 2026, including those of gpt-oss and Qwen3.5, under which style overlap rose, so the decline of *delve* need not mean that writing stopped moving. Detectors, prevalence estimators and marker lists built on a fixed set of models may therefore understate recent change unless the reference is refreshed. Our data cannot fully say why abstracts in 2026 resemble the newest models. The 2026 open models peaked before or as they were released, so authors' use of them cannot explain their peaks; the models may have learned from recent writing, or they and the authors may share the style of the commercial models in use, which we did not run. Either way, LLM style and human scientific style can no longer serve as independent references for each other, a precondition for the recursive dynamics studied in model-collapse work (Shumailov et al., 2024; Gerstgrasser et al., 2024) but not evidence of them.

8 Conclusion

In arXiv abstracts, Machine-Style Overlap rose from 9.2% before ChatGPT to 40.6% in 2025, well above its pre-ChatGPT trend. Part of this shift is towards generators whose pretraining data predate ChatGPT. Overlap with a single generator tended to peak later for more recent generators, so studies of LLM influence on writing should report the vintage of their reference models and update it.

Limitations

MSO is defined relative to a feature map and a researcher-chosen generator set, and its trajectory after 2025 depends on that set's vintage (§6.4). Four generators are from 2026; one of them was run without decoding control in batches of 50 prompts and is the system that drafted this paper. The pre-ChatGPT-data region rests on four generators, three of them built on Llama 2, and their instruction tuning used data from 2023. The vintage comparison holds the prompt fixed but not decoding (for RAID) or output length: medians range from 102 to 110 words for Vicuna, Llama-2-Chat 7B and GPT-4, whose outputs are the shortest, and from 138 to 186 for the 2025–2026 open models. The three 2026 open models were added after the pattern of the other generators had been seen, and single-generator regions rest on 150 texts each (up to 450 for RAID models), so instrument precision differs by vintage. Cross-validation grouped the 2026 model's outputs by source document, not by session, which may flatter its held-in recall. Mined marker lists contain some topic and register words. Post-ChatGPT "human" documents may be partly LLM-written, so MSO cannot separate LLM use from human adoption of LLM style, and no causal claim is made. The overall arXiv series averages a stratified sample of eight fields (two of them in physics) and is not the arXiv population. Changes in authorship, subfields and submission volume may contribute despite stratified sampling and topic and length checks, and excluding papers revised after ChatGPT makes the late pre-period sample more selected. Confidence intervals ignore instrument variation and clustering by field-month. Classifier families agree on population-level MSO for abstracts (22.8% vs 22.6%) but less on individual documents (Cohen's $\kappa = 0.64$; speech 0.26, essays 0.29), and a perplexity instrument reverses its bias between two learner populations, so no score here should be used to judge an individual writer. All text is English and truncated to 250 words. Parliamentary speech differs from abstracts in many ways, its linear trend fails its own placebo tests, and the 2022 change of government altered the speaker population six months before ChatGPT. The dispersion measure depends on document length. The learner analysis is cross-sectional, concerns US school learners, and rests on 15 in-region learner essays; its perplexity instruments use two small language models and thresholds calibrated on PERSUADE. The analysis plan was committed to our own repository and never frozen with a third party.

Ethics Statement

The study uses only public datasets under their licences: arXiv metadata, including titles and abstracts (CC0); Australian Hansard (Parliament of Australia, CC BY-NC-ND 4.0), through the database of Katz and Alexander (2025) (CC BY 4.0); PERSUADE 2.0 (CC BY-NC-SA 4.0, whose share-alike terms apply to derived data we release); and RAID and the Hugging Face release of Liang et al. (2023) (MIT). Their US 8th-grade essays come from the authors' GitHub repository, which has no licence file but whose README shows an MIT badge, and derive from the Hewlett Foundation ASAP data, so we release only aggregate results for them. Because Hansard text may not be adapted, we do not release the LLM revisions of speeches (P2). The study involves no human participants. Hansard speakers are members of parliament speaking in public office; we release only speech and speaker identifiers, party and opening sentences, which are already public. PERSUADE essays were de-identified by their creators, the released LLM revisions of them keep that masking, and we release no human-written essays. The local models are used under their licences (Llama 2 and Llama 3.1 community licences, which also cover Vicuna; Gemma Terms of Use for Gemma 3; Apache 2.0 or MIT for the others), all of which permit this research use; some restrict training other models on their outputs, which the data release notes. All local generation (11,730 texts from models of 7 to 20 billion parameters) and all analyses ran on one desktop PC with a consumer GPU (AMD Radeon RX 7800 XT, 16 GB) over two days; GPU-hours were not logged. The learner analysis compares groups, never individuals, and we caution against using any overlap or detector score to judge an individual writer, or using MSO as a detector.

Acknowledgements: AI Use and Authorship

An AI system (Claude Opus 5.5, Anthropic, used as an agent through Claude Code) carried out most of this study. The human author chose the topic, the goal of a fully disclosed AI-conducted study, and the constraints (public data only, no

human participants, local compute), and approved the analysis plan in the working conversation; deviations (Appendix A) were made by the AI system and reported to the author. The AI system proposed the research question, the term "Machine-Style Overlap" and the hypotheses, wrote the analysis plan and all code, generated the local-model texts, ran the analyses, made the figures and drafted all text; no human has edited the text. The same system produced the 2026-model texts (§4), so its own outputs are part of the reference set. Simulated peer review by separate sessions of the same model family, which share its blind spots, informed revisions. Separate AI sessions checked every citing sentence against the cited work's abstract or full text in two full passes, some sources only through metadata or mirrors. The checks led to rewordings, one removed citation and the replacement of preprints by published versions. Citations were not checked by a human; the human author will read the final paper before release. Every numerical result of our analyses in the paper is inserted automatically from the analysis outputs; a few counts and orderings stated in words were read from the same outputs. Full responsibility for the content, including the citations and the analysis choices, lies with the human author. No AI system is listed as an author.

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A Deviations from the analysis plan

The plan was committed to version control before any result was computed (commit `ef4fa21`); it was never frozen with a third party. (1) arXiv was sampled through the API with a stratified design instead of full harvesting; the first sampler truncated large windows and was replaced (only the second is used); pre-ChatGPT papers revised after ChatGPT are excluded. (2) PERSUADE 2.0 replaced ELLIPSE, which could not be obtained; the essays of Liang et al. (2023) (TOEFL, US 8th-grade and US college) are a different genre and are used only for out-of-genre checks: the perplexity replication and the style check below. The style instrument places 34.1% of TOEFL, 5.7% of 8th-grade and 51.4% of college essays in the PERSUADE-defined region, which shows how genre-specific the region is. (3) Phi-4, Granite and the 2026 model were chosen after an arXiv pilot on partial data and before the first review. The 2026 model's texts were merged two minutes after a preliminary arXiv run without them was committed; Phi-4 and Granite were first used after the first review. The pilot may have informed this decision. Before any Hansard result, a bug in the speech identifiers was fixed (the corpus field we first used as a speech identifier identifies the speaker); an essay pilot on partial generations changed no analysis choice. RAID enters the main region only through its chat models, and each generator is capped at 450 training texts. (4) Added after the first review: LaTeX and encoding normalisation, grouped calibration, generator-set, drift, math, topic, length and dash analyses, the proficiency adjustment, the perplexity instrument and the dispersion measure. (5) Added after the second review: the vintage analysis with Llama-2-Chat 7B and Vicuna 7B; the three 2026 open models (Qwen3.5, Gemma 4, Granite 4.2), added after the single-generator pattern of the

other 16 generators had been seen; mined marker words, the placebo grid, pre-trend variants, block bootstrap, further drift origins, Holm correction and the computer science vs mathematics contrast, length-fixed dispersion and GPT-2 perplexity, the event study and the implied-prevalence calculation; a cleaning fix (whole "Title:" lines in LLM outputs are dropped), after which every analysis was re-run. (6) Added after the third review: a second cleaning fix (keyword lines, word-count notes and inline labels), again with a full re-run; quarterly single-generator series, seed ranges and intervals for the 2026 changes, drift placebos for the pre-ChatGPT-data region, a 30-month placebo grid, a never-revised check, feature-family ablations, field-specific detection rates and a zero-shot detector baseline. The rank correlation of release date with the 2025-to-2026 change was computed before the one with the peak quarter. Added after a pre-release methods audit: summary-style RAID outputs are also dropped from the single-generator regions (they had been dropped only from the pre-ChatGPT-data region), with the vintage analyses re-run; the post-trend fit to June 2025, fixed field weights, the document bootstrap of peak quarters, generator-side subsamples and the cutoff and marker-peak comparisons. (7) The self-test of the paper's own text and the plan's arXiv embedding instrument were dropped. (8) The plan named cs.CL for one comparison; all of cs is used. The June 2025 evaluation month was fixed in code, not in the plan. Hansard was restricted to 2015–2025. H5 was not supported.

B Top features

Table 3 lists the ten largest positive and negative coefficients of the primary arXiv instrument.

LLM-typical	coef.	Human-typical	coef.
MTLD lexical diversity	1.30	dependency conj	−0.49
POS trigram DET_NOUN_VERB	0.60	dependency amod	−0.48
punctuation ,	0.53	word *by*	−0.44
word *that*	0.45	sentence length (mean)	−0.42
dependency det	0.39	word *are*	−0.41
dependency advcl	0.36	dependency aux	−0.41
POS trigram DET_VERB_NOUN	0.33	word *is*	−0.39
POS trigram NOUN_VERB_SCONJ	0.32	dependency ROOT	−0.35
POS trigram PUNCT_PRON_NOUN	0.32	dependency advmod	−0.34
word *these*	0.30	dependency compound	−0.33

Table 3: Largest standardised coefficients of the primary arXiv instrument (logistic regression, 14 generators).

C Held-out-generator recall

Held-out generator	Recall (%)	arXiv MSO (%)
Claude Opus 5.5	20.7	20.6
gpt-oss 20B	65.6	20.8
Qwen3 8B	76.4	22.8
Llama 3.1 8B	83.1	22.6
Mistral 7B	87.8	23.4
Phi-4 14B	88.9	22.7
Gemma 3 12B	90.0	23.1
Granite 3.3 8B	91.3	22.9
RAID Cohere-chat	59.6	23.4
RAID Mistral-chat	83.9	23.1
RAID Llama-chat	91.1	22.8
RAID ChatGPT	93.3	23.1
RAID MPT-chat	94.6	23.0
RAID GPT-4	95.6	23.3

Table 4: Recall on each generator when it is held out of training, and the resulting MSO of all scored arXiv abstracts (2015–2026; 22.8% with all 14 generators).

D Single-generator peaks

Table 5 gives each single-generator region's peak quarter with its bootstrap range, and the stability of its 2025-to-2026 change.

E Field results

Figure 3 shows the pre-ChatGPT baseline and the June 2025 trend gap for each field.

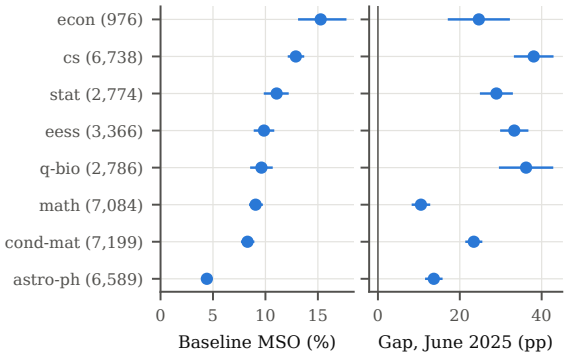


Figure 3: Left: pre-ChatGPT MSO by arXiv field (n of pre-ChatGPT scored abstracts). Right: trend gap in June 2025. Lines are 95% intervals. The baseline does not predict the gap.

F Parliamentary speech

The Hansard trend gap is small and not significant at the primary threshold (1.3 pp, CI −0.7 to 3.4; Figure 1b), about 0.2 times its extrapolated trend value against 2.1 for abstracts (0.21 against 1.47 on the logit scale). With the arXiv region rebuilt from the eight generators used for Hansard

Generator	Peak	80% range	Draws	2025→26	Same sign
GPT-4 (RAID)	2025Q3	2025Q1–2025Q4	33.8	−3.6	10/10
ChatGPT (RAID)	2025Q1	2025Q1–2025Q2	81.7	−5.1	10/10
MPT-30B-chat	2025Q1	2025Q1–2025Q2	71.4	−3.6	10/10
Llama-2-Chat 70B	2025Q2	2025Q1–2025Q2	59.5	−5.5	10/10
Llama-2-Chat 7B	2025Q1	2024Q3–2025Q2	69.7	−5.2	10/10
Vicuna 7B v1.5	2024Q3	2024Q3–2025Q1	71.1	−2.6	10/10
Mistral-7B-Instr. v0.1	2025Q1	2025Q1–2025Q3	75.7	−5.4	10/10
Cohere command	2023Q2	2023Q2–2024Q3	54.9	−3.0	10/10
Mistral 7B v0.3	2025Q3	2025Q2–2025Q4	49.8	−4.6	10/10
Llama 3.1 8B	2025Q2	2025Q1–2025Q3	44.8	−5.0	10/10
Phi-4 14B	2025Q2	2025Q2–2025Q3	73.8	−4.3	10/10
Gemma 3 12B	2025Q4	2025Q3–2026Q1	46.5	−2.0	10/10
Granite 3.3 8B	2025Q4	2025Q3–2025Q4	60.6	−2.6	10/10
Qwen3 8B	2025Q4	2025Q2–2025Q4	42.8	−3.7	10/10
gpt-oss 20B	2026Q2	2026Q2–2026Q2	94.4	9.0	10/10
Qwen3.5 9B	2026Q1	2026Q1–2026Q2	80.6	2.4	10/10
Gemma 4 12B	2025Q4	2025Q4–2026Q1	67.8	−0.9	10/10
Granite 4.2 8B	2026Q2	2026Q1–2026Q2	76.8	7.4	10/10
2026 model	2026Q3	2026Q3–2026Q3	91.6	9.9	10/10

Table 5: Quarter of peak MSO with the region defined by one generator (title-only task; generators in order of release). 80% range and Draws: 10th–90th percentile of the peak quarter and share (%) of draws in which the observed peak recurs, over 1,000 document bootstraps within quarters. 2025→26: annual change (pp); Same sign: refits on 80% subsamples of the generator's texts whose change has the same sign.

(arXiv gap 32.1 pp), the difference between the genres is 30.7 pp (CI 28.3–33.2). Hansard is a weak comparison, however. Its gap is 1.4 pp at threshold 0.6 (CI −0.1 to 2.9), and its nonparametric gap ranges from −3.2 to 3.1 pp across pre-trend specifications. Of its placebo breaks, 23.5% give $p < 0.05$, and the 2022 change of government altered the speaker population six months before ChatGPT. Published marker words also rose in speeches (1.6 pp above trend). We read Hansard as evidence that parliamentary speech changed much less than abstracts.

G Dispersion

To connect overlap to homogenisation we measured, for each year, the mean Euclidean distance of abstracts to the year's centroid in the feature space standardised on pre-ChatGPT abstracts. On all abstracts, dispersion falls steadily from 34.20 in 2015 to 30.86 in 2026, but abstracts also grew longer, and per-token rates are less noisy in longer texts. With every abstract cut to its first 150 words (abstracts of at least 150 words, 1,200 per year), dispersion declines slowly before ChatGPT (−0.11 per year) and in 2025 lies 0.05 above that trend (bootstrap CI −0.36 to 0.51). The interval excludes an extra decline larger than about three years of the pre-ChatGPT trend, so any faster loss of stylistic spread is small next to the shift towards the LLM region. Diversity of content or ideas may behave differently (Padmakumar and He, 2024; Doshi and Hauser, 2024).

H Learner essays

Our analysis plan predicted that essays by English-language learners would show *higher* MSO. They show lower: 2.3% vs 4.2% (15 of 666 learner essays in the region; odds ratio adjusted for prompt and grade 0.55, CI 0.32–0.95; lower, and significant at $p < 0.05$, in all 10 training draws). Most of the difference is proficiency: adding the holistic essay score reduces the odds ratio to 0.87 (CI 0.50–1.52) and the continuous classifier margin from −0.37 to −0.12 standard deviations (CI −0.21 to −0.03). In essays the LLM region is largely a region of fluent, lexically diverse writing. The other instruments agree in direction (Figure 4). Sentence embeddings flag 5.3% of learner and 10.0% of other essays. Perplexity under GPT-2 separates human from LLM essays less well than style features (AUC 0.830 against 0.988), so it flags 29.2% of all human essays. It flags 15.5% of learner vs 30.6% of other essays (odds ratio 0.47; 0.66, CI 0.52–0.85, after adjusting for proficiency), because these learners' essays are *less* predictable (mean token log-perplexity 3.74 vs 3.43). The same GPT-2 instrument reproduces the bias reported by Liang et al. (2023) on their data: it flags 63.7% of TOEFL essays by non-native test-takers, which are highly predictable (3.07), but 8.0% of

US 8th-grade essays (3.72); Qwen2.5-0.5B gives the same pattern (41.8% vs 3.4%). GPT-2's training data predate the public release of the TOEFL essays (and PERSUADE), so that contrast is unlikely to come from memorised test essays, although the 8th-grade essays derive from data public since 2012. In these data a perplexity instrument penalises predictable writing. The TOEFL essays are predictable and the PERSUADE learners' essays are not, so which learners such a detector flags depends on the population, consistent with detectors that showed no bias in a large writing assessment (Jiang et al., 2024). The TOEFL and 8th-grade sets also differ in genre, prompt, age and length.

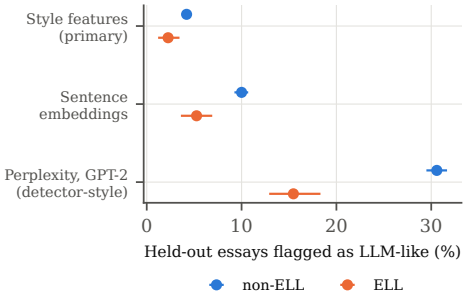


Figure 4: Share of held-out pre-ChatGPT student essays flagged by three instruments, by ELL status (ELL $n = 666$, non-ELL $n = 6,758$; lines: 95% bootstrap intervals). Levels differ across instruments; all flag learner essays less often.